MTH303 JAN 2021 Exam With Solution

Principles of Graph Theory (Singapore University of Social Sciences)

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MTH303

Timed Online Assignment (TOA)
Jan 21 semester
Guided Solution

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Question 1 (Total: 25 marks)

An undirected graph G is given as follows in Figure Q1:

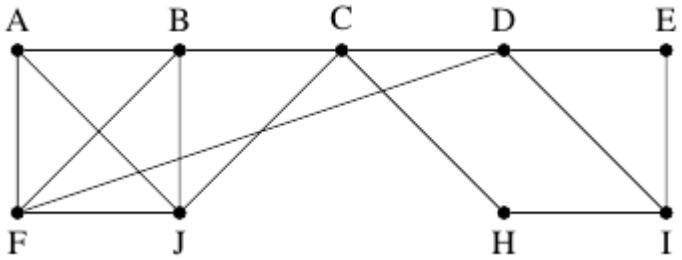


Figure Q1

(a) Draw and label the spanning tree of G obtained by using a depth-first search in which the vertices are traversed in alphabetical order starting from vertex A.

(4 marks)

(b) Construct an Eulerian trail as well as a Hamiltonian cycle in G.

(8 marks)

(c) Provide an edge-colouring of G using at most 5 colours.

(4 marks)

(d) Provide a maximal set of vertex-disjoint AD-paths in G and compute the vertex connectivity $\kappa(G)$.

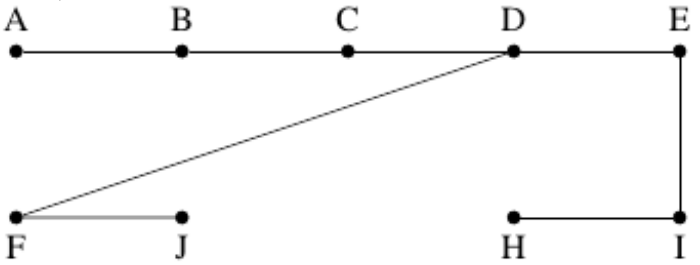
(5 marks)

(e) Provide a minimum dominating set of G and calculate the domination number $\mathrm{dom}(G)$.

(4 marks)

Marking Guide:

(a) The spanning tree obtained using depth-first search, alphabetical order, is:

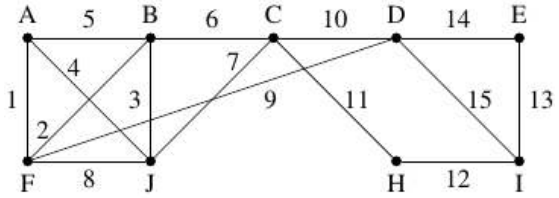


NOTE: There is only one correct answer.

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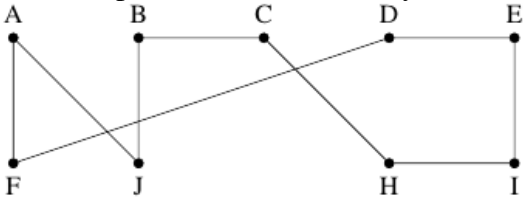
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(b) An example of an Eulerian trail is:



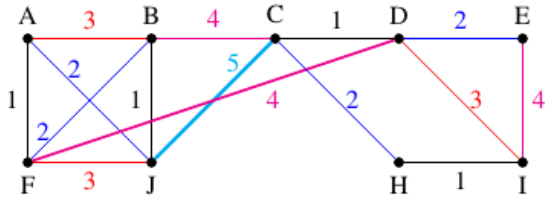
The trail is A-F-B-J-A-B-C-J-F-D-C-H-I-E-D-I.

An example of a Hamiltonian cycle is :



The cycle is A-F-D-E-I-H-C-B-J-A.

(c)



One possible edge colouring is given by the above.
Here, we colour edges AF, BJ, CD and HI with colour 1. Colour edges AJ, BF, CH and DE with colour 2. Colour edges AB, FJ and DI with colour 3. Colour edges DF and EI with colour 4. Colour edge CJ with colour 5.
Any edge colouring using 5 colours is acceptable (note that 4 colours is impossible for the above graph).

(d) A maximal set of vertex-disjoint AD-paths is { ABCD, AFD }

Since G is a simple graph that contains a cycle, removing any single vertex cannot disconnect G. On the other hand, removal of vertices C,D will disconnect G. Hence, $\kappa(G) = 2$.

(e) An example of a minimum dominating set of G is { B, I }.

The domination number $\mathrm{dom}(G)$.is 2.

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Question 2 (Total: 15 marks)

An edge-weighted graph G is given as follows in Figure Q2:

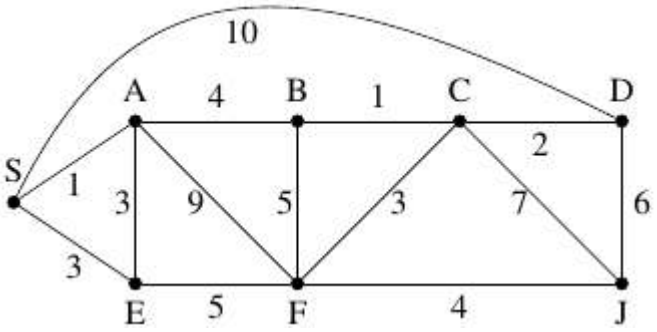


Figure Q2

(a) Apply Dijkstra's algorithm to determine the weight of a shortest path from S to each of the other vertices. Give the labels and predecessors of each vertex at every step of the algorithm.

(12 marks)

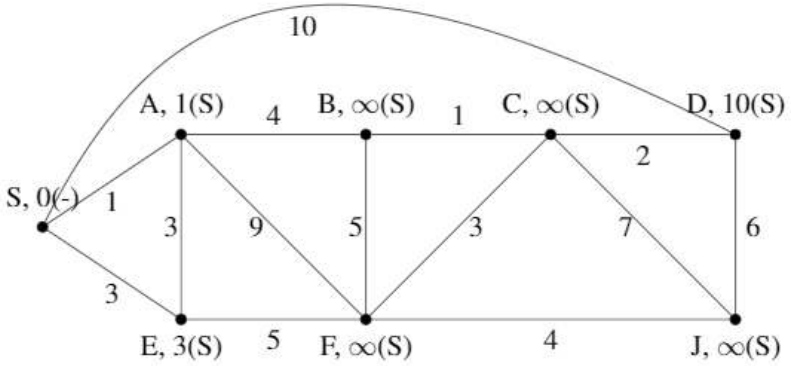
(b) Construct a shortest path from S to J.

(3 marks)

Marking Guide:

(a) The labels and predecessors at each stage are indicated below on the respective graphs. In the working below, **P** denotes the permanent set.

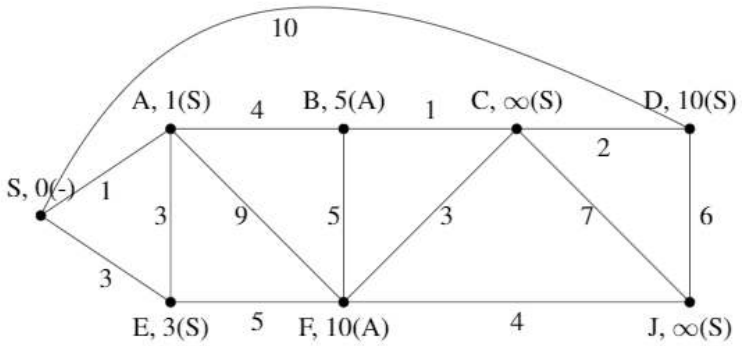
$\mathbf{P} = \{S\}$



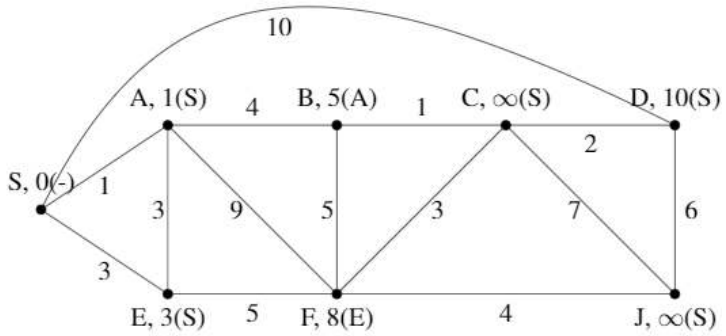
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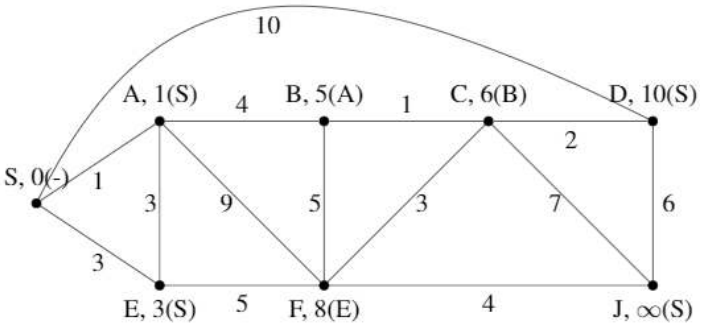
P = $\{S, A\}$



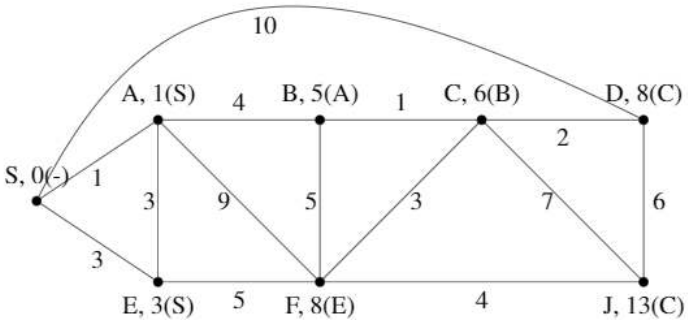
P = $\{S, A, E\}$



P = $\{S, A, E, B\}$



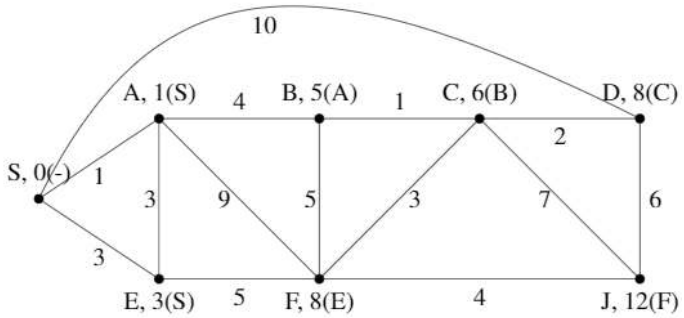
P = $\{S, A, E, B, C\}$



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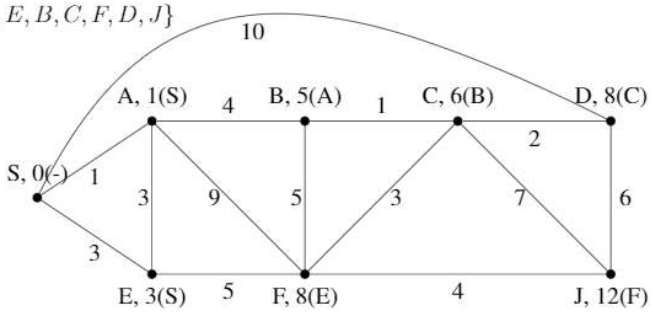
$\mathbf{P} = \{S, A, E, B, C, F\}$



The addition of the final two vertices D and J produce no change in the existing labels or predecessors.

$\mathbf{P} = \{S, A, E, B, C, F, D\}$

$\mathbf{P} = \{S, A, E, B, C, F, D, J\}$



The final labels indicate the minimum distance from S to each of the vertices.

(b) A shortest path S to J is given by S-E-F-J.

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Question 3 (Total: 15 marks)

A basic network G with source S and sink T is given as follows in Figure Q3:

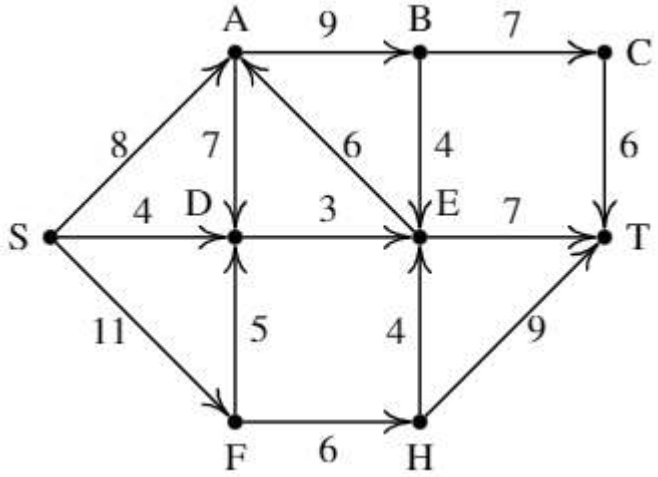


Figure Q3

(a) Use the Flow Augmentation Algorithm to determine the maximum flow. Give the flow-augmenting path and the value of the flow along each arc at every step of the algorithm.

(8 marks)

(b) Give a cut whose capacity is equal to the maximum flow.

(3 marks)

(c) Determine with justification how the maximum flow would change if the capacity of the arc DE is increased to 6.

(4 marks)

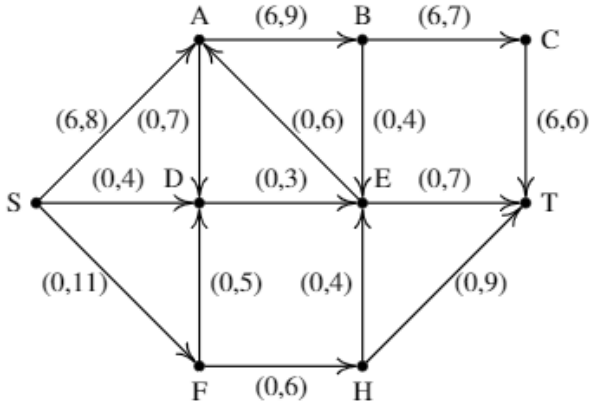
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Marking Guide:

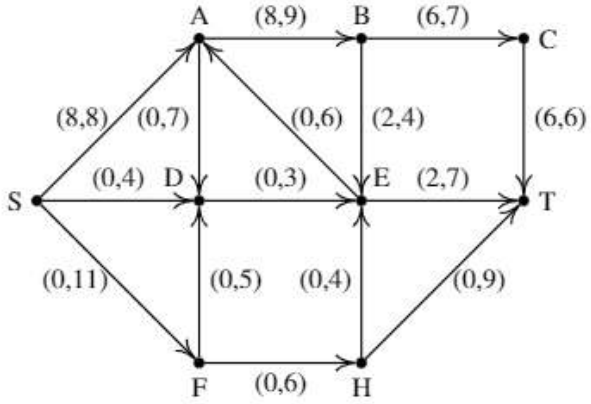
(a) First iteration:

Augmenting path: SA, AB, BC, CT. Increase flow by 6 units.



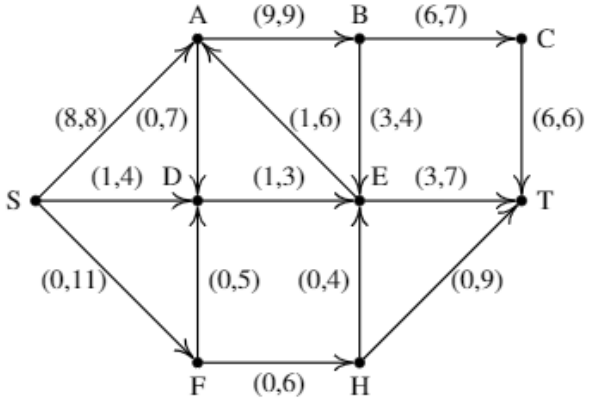
Second iteration:

Augmenting path: SA, AB, BE, ET. Increase flow by 2 units.



Third iteration:

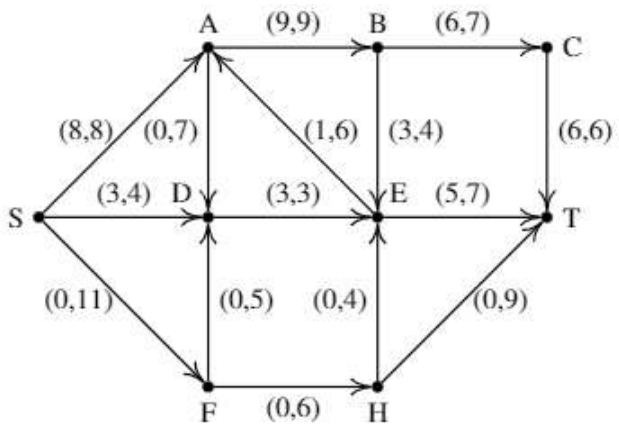
Augmenting path: SD, DE, EA, AB, BE, ET. Increase flow by 1 unit.



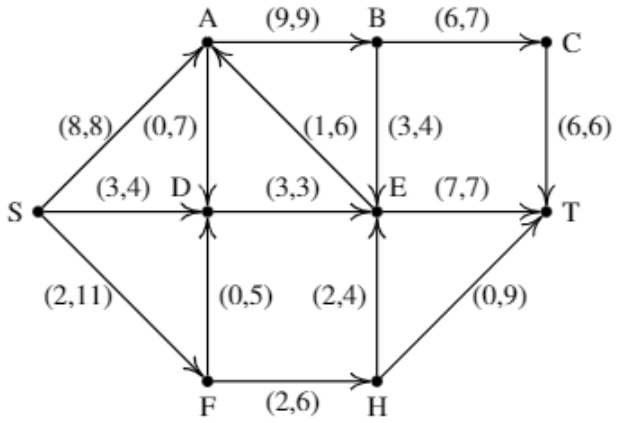
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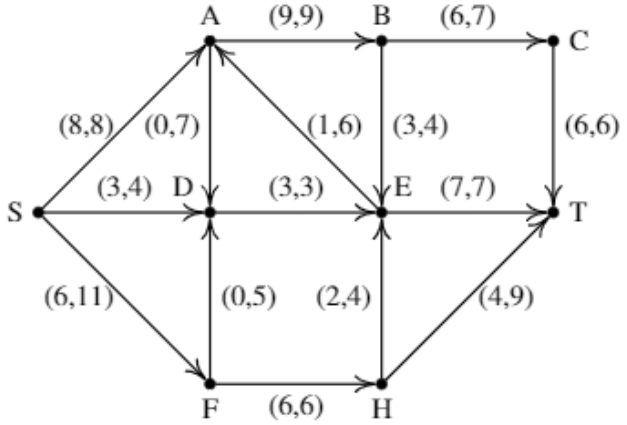
Fourth iteration:
Augmenting path: SD, DE, ET. Increase flow by 2 units.



Fifth iteration:
Augmenting path: SF, FH, HE, ET. Increase flow by 2 units.



Sixth iteration:
Augmenting path: SF, FH, HT. Increase flow by 4 units.

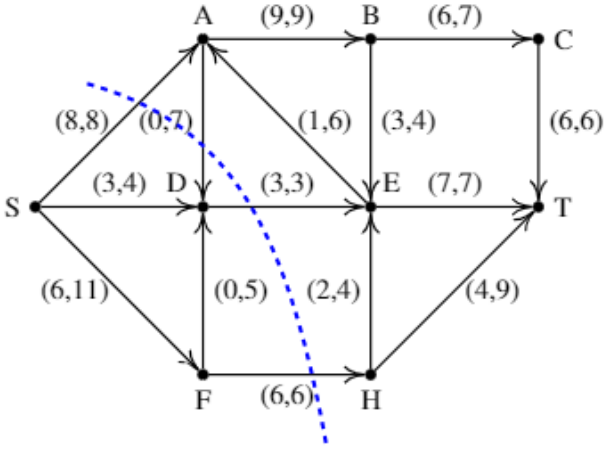


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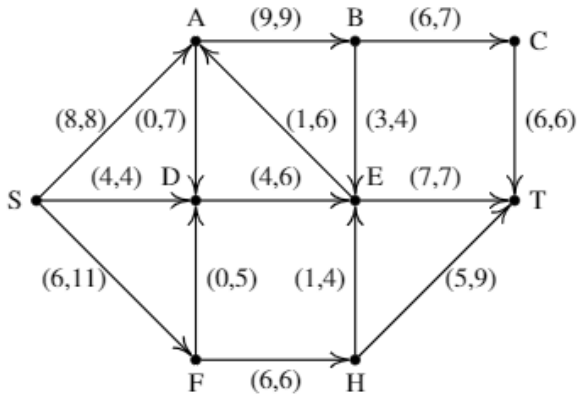
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Since there are no more flow-augmenting ST paths, the maximum flow has been achieved. The value is $8 + 3 + 6 = 17$, the sum of flow of the arcs incident from S. This is the same as the sum of flow of arcs incident to T $(6 + 7 + 4)$, and also the same as the sum of the flow increases with each iteration $(6 + 2 + 1 + 2 + 2 + 4)$.

(b) { SA, AD, DE, FH } is a cut whose capacity (the capacity of a cut is defined as the sum of capacities of the forward arcs) is $8 + 3 + 6 = 17$, the same value as the maximum flow.



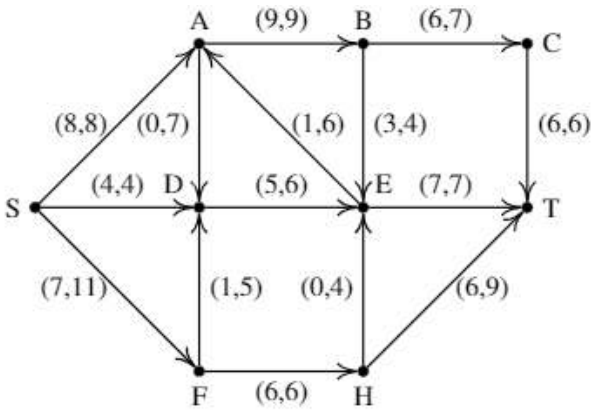
(c) Suppose the capacity of DE increases from 3 to 6.
By making use of the augmenting path SD, DE, EH (reverse arc), HT, we can increase the flow by 1 unit.



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Finally, by making use of the augmenting path SF, FD, DE EH (reverse arc), HT, we can increase the flow by 1 unit, making a total of 2 units.



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Question 4 (Total: 10 marks)

An image is displayed on an 8x8 screen, as shown in Figure Q4 below.

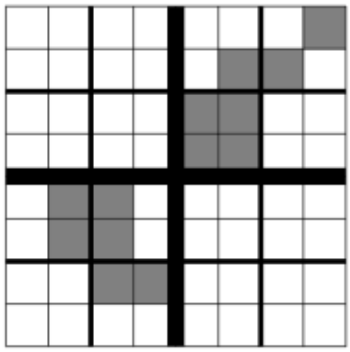


Figure Q4

(a) Draw the corresponding pruned quad tree of the image.

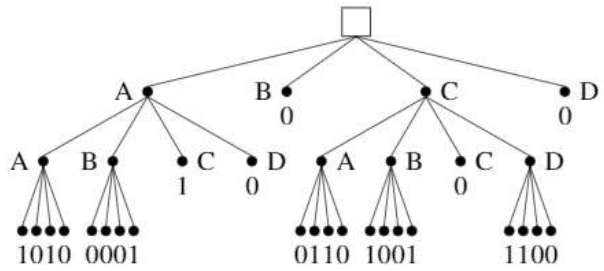
(6 marks)

(b) Draw the quad tree that results from reflecting the image about the line going from the top left to the bottom right corner of the screen.

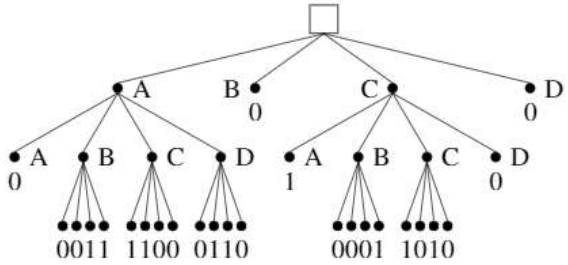
(4 marks)

Marking Guide:

(a)



(b)



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Question 5 (Total: 15 marks)

Figure Q5 shows the cost matrix of a weighted bipartite matching problem.

	y_1	y_2	y_3
x_1	5	3	8
x_2	2	7	3
x_3	9	4	8

Figure Q5

(a) Employ the Hungarian Algorithm to construct a maximum matching with minimum total cost. Show your working in full, including the cost matrices, partial graphs, and matching used at each step of the algorithm.

(8 marks)

(b) It is known that the Hungarian Algorithm applied to a general $n \times n$ cost matrix is of worst-case time complexity $O(n^3)$. Show that there exists an algorithm that can solve the weighted bipartite matching problem for a general $n \times m$ cost matrix with worst-case time complexity $O(\max(n^3, m^3))$, where $\max(a, b)$ refers to the maximum of the two numbers a, b.

(7 marks)

Marking Guide:

(a) Let $X = \{x_1, x_2, x_3\}$ and let $Y = \{y_1, y_2, y_3\}$.
For each vertex in X, assign the lowest cost in the row to the vertex.

		y_1	y_2	y_3
3	x_1	5	3	8
2	x_2	2	7	3
4	x_3	9	4	8

Decrease the weights of edges by the weights on the X-vertices to which they are incident. Now, for each vertex in Y, assign the lowest cost in the column to the vertex.

		0	0	1
		y_1	y_2	y_3
3	x_1	2	0	5
2	x_2	0	5	1
4	x_3	5	0	4

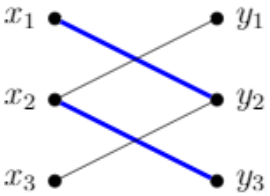
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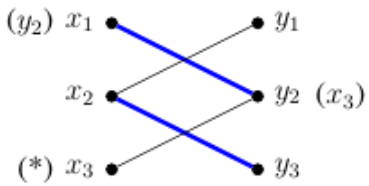
Then decrease the weights of edges by the weights on the Y-vertices to which they are incident.

		0	0	1
		y_1	y_2	y_3
3	x_1	2	0	4
2	x_2	0	5	0
4	x_3	5	0	3

Construct a partial graph consisting of the vertices of the original bipartite graph, together with only those edges whose current cost is 0. An initial matching in the partial graph is also given in **blue**.

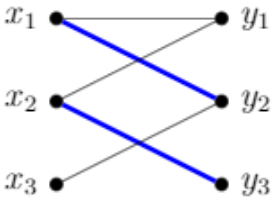


Label the X and Y vertices as prescribed by the Hungarian Algorithm.

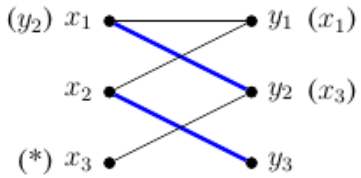


Modify the vertex weights according to the Algorithm, update the current cost matrix, and then construct the updated partial graph.

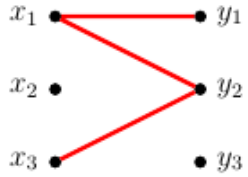
		0	−2	1
		y_1	y_2	y_3
5	x_1	0	0	2
2	x_2	0	7	0
6	x_3	3	0	1



Again, label the X and Y vertices as prescribed by the Hungarian Algorithm.



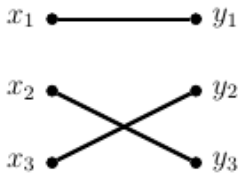
Since a breakthrough at vertex y_1 has occurred, we trace back to obtain the augmenting path $M = y_1x_1y_2x_3$



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and thereby obtain the required optimum matching, which has total cost $5 + 3 + 4 = 12$.



(b) For a general $n \times m$ cost matrix, suppose without loss of generality that $m > n$.

Create an $m \times m$ cost matrix by adding extra rows (corresponding to dummy vertices) to the original cost matrix in which the entries of the extra rows are each larger than all the entries in the original cost matrix.

The Hungarian Algorithm can then be applied to solve the new cost matrix, with time complexity $O(m^3)$. After the algorithm is complete, remove the dummy vertices and their incident edges from the derived matching to obtain an maximum matching with minimum cost of the original assignment problem. All the extra steps only require time of order $O(\max(m, n))$, and so the worst-case time is $O(m^3)$.

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Question 6 (Total: 20 marks)

Let G be a simple edge-weighted graph with no isolated vertices. We call a set of edges in G acyclic if no cycle is contained in the subgraph of G spanned by those edges.

(a) Let X, Y be two sets of acyclic edges in G such that $|X| > |Y|$, where $|S|$ denotes the number of elements in the set S. Show that there exists an edge $e \in X$ not belonging to Y such that $Y \cup \{e\}$ is a set of acyclic edges in G.

(7 marks)

(b) A greedy algorithm is applied in which the edges of G are chosen one at a time. At each step of the algorithm, an edge of smallest weight is chosen such that no cycle is formed. Prove that this algorithm will always succeed in producing a minimum spanning forest of G.

(8 marks)

(c) A student suggests using a greedy algorithm to solve the maximum matching problem in a weighted bipartite graph. For a weighted bipartite graph H, select edges of minimum weight in H one at a time subject to the condition that the edges remain a matching after each selection. Demonstrate with a suitable example that this greedy algorithm need not always produce a maximum matching with minimum cost in H.

(5 marks)

Marking Guide:

(a) Let G_X be the subgraph of G consisting of the vertices of G together with only the edges in X, and let G_Y be the subgraph of G consisting of the vertices of G together with only the edges in Y.

Since $|X| > |Y|$, G_Y has more connected components than G_X, and so, there is a connected component C of G_X that contains vertices belonging to different connected components of G_Y.

It follows that we can find an edge e from the component C of G_X such that the end points of e belong to two different connected components of G_Y.

Since the edge e does not belong to Y, and e connects vertices of different components of G_Y, the set of edges $Y \cup \{e\}$ is an acyclic set of edges in G.

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(b) Let X_m denote the set of edges chosen after the mth iteration of the greedy algorithm. Since X_m is acyclic at every step of the algorithm, a spanning forest T is obtained when the algorithm terminates (no more choice of edges are possible). We prove that the spanning forest T is a minimum one.

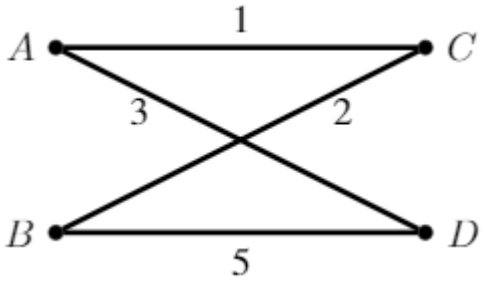
Suppose not. Then there exists a spanning forest S of smaller total weight than T. We can order the edges of S from smallest weight to largest weight. Let Y_m be the set comprising of the first m edges of S, and as in the previous paragraph, let X_m continue to denote the set of the first m edges of T. Since S is of smaller total weight than T, there is a smallest integer k such that the kth edge of S is of strictly smaller weight than the kth edge of T.

Let f denote the kth edge of T, so that $X_{k-1} \cup \{f\} = X_k$.

Since $|X_{k-1}| < |Y_k|$ and both sets of edges are acyclic, by part (a), there exists some edge e of Y_k such that e is not a member of X_{k-1} and the set of edges $X_{k-1} \cup \{e\}$ is acyclic. But since the kth edge of S is of strictly smaller weight than the kth edge of T (namely, f), the weight of e must be strictly smaller than the weight of f. This leads to a contradiction, because the greedy algorithm that produced T would have either added e or some other edge of smaller weight than e at the kth step, rather than edge f.

It follows therefore that the algorithm that produced T must have resulted in a minimum spanning forest.

(c) Consider the weighted bipartite graph shown below.



The greedy algorithm proposed by the student would have chosen edges AC and BD, giving a maximum matching of total weight $1 + 5 = 6$.

However, the maximum matching of minimum weight comprises of edges AD and BC, of total weight $2 + 3 = 5$.

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Hence this example shows that the greedy algorithm need not produce a maximum matching of minimum weight.

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